

**A Logistic Regression approach to  
predicting insurance claim denials before  
submission — enabling proactive revenue  
protection.**

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## NovaCare Insurance

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## 1. Project Overview

NovaCare Insurance is experiencing a persistent revenue leakage problem: 39.58% of all submitted claims are being denied, resulting in \$7.22M in lost revenue annually. The billing team only discovers a denial after the claim has been rejected — by which time the damage is done and the team must appeal, resubmit, or write off the loss.

This project, led by Team 3 of DataVerseAfrica Cohort 4.0, addresses this problem with a predictive model. By learning from 6,150 historical claims, the model assigns every new claim a denial probability score between 0 and 1. Claims scoring above 0.5 are flagged as high-risk for pre-submission review — giving the billing team the opportunity to fix problems before they become denials.

### Project Stages

| Stage   | Description                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------------------|
| Stage 1 | Data cleaning and standardisation in SQL — 6,150 rows, inconsistent provider names, department formats and casing issues resolved. |
| Stage 2 | Denial pattern analysis in Python — identifying top denial reasons, high-risk departments, providers, and payer types.             |
| Stage 3 | Predictive modelling — Logistic Regression trained on historical claims to score future submissions.                               |
| Stage 4 | Power BI dashboard — four-page interactive dashboard presenting problem scale, concentration, cost, and model predictions.         |

## 2. Dataset

|                              |                               |                              |                               |                |
|------------------------------|-------------------------------|------------------------------|-------------------------------|----------------|
| <b>6,150</b><br>Total Claims | <b>2,434</b><br>Denied Claims | <b>39.58%</b><br>Denial Rate | <b>1,230</b><br>Test Set Size | Train Set Size |
|------------------------------|-------------------------------|------------------------------|-------------------------------|----------------|

The dataset contains 6,150 insurance claims spanning all 8 departments at NovaCare. The 80/20 train-test split was stratified — ensuring both the training set (4,920 claims) and test set (1,230 claims) contain the same 39.58% denial rate, giving an honest evaluation of model performance on unseen data.

## 3. Feature Strategy — Columns Kept, Dropped, and Created

### Columns Kept

| Column        | Reason Kept                                                                   |
|---------------|-------------------------------------------------------------------------------|
| department    | Core predictor — denial rates vary significantly across departments (38%–41%) |
| payer_type    | Government and Self-Pay payers showed 40%+ denial rates in pattern analysis   |
| icd10_primary | Diagnosis code — drives medical necessity review decisions                    |

| Column                 | Reason Kept                                                                          |
|------------------------|--------------------------------------------------------------------------------------|
| icd10_description      | Human-readable diagnosis label — adds interpretable clinical context (see Section 4) |
| cpt_procedure          | Procedure code — links to billing rules and coverage limits                          |
| cpt_description        | Procedure description — adds label context for encoding                              |
| provider_name          | Provider identity — Dr. Yetunde Okafor had 70.59% denial rate                        |
| billed_amount          | Higher-value claims face greater insurer scrutiny                                    |
| risk_score             | Pre-existing composite risk flag from the claims system                              |
| claim_processing_days  | Processing lag may correlate with documentation gaps                                 |
| days_to_submit         | Late submission directly causes Timely Filing denials (\$0.89M leakage)              |
| length_of_stay_days    | Longer stays involve more complex billing with higher denial risk                    |
| is_duplicate_claim     | Duplicate submissions are denied at 92% — the model's strongest signal               |
| claim_before_discharge | Premature claims lack complete documentation — higher denial risk                    |
| risk_payer             | Payer-level risk flag — encodes known high-risk payer behaviour                      |
| risk_processing        | Processing risk flag from claims intake system                                       |

## Columns Dropped — and Why

A critical rule in predictive modelling is avoiding data leakage — including information in the model that would only be available AFTER the outcome has occurred. A model trained on leaked data learns to cheat: it appears highly accurate in testing but fails completely on real future claims. The following columns were removed for this reason or because they carry no predictive signal:

| Column Dropped | Reason                                                                  |
|----------------|-------------------------------------------------------------------------|
| claim_id       | Unique identifier — no predictive pattern, would cause overfitting      |
| patient_id     | Unique identifier — individual patients do not generalise               |
| encounter_id   | Unique identifier — no signal                                           |
| provider_id    | Replaced by provider_name which carries the signal                      |
| claim_status   | DIRECT SOURCE of target variable — including it is the ultimate leakage |
| denial_reason  | Only exists after a denial occurs — not available at submission time    |
| allowed_amount | Set by insurer post-adjudication — severe post-event leakage            |
| paid_amount    | Only known after payment decision — post-event leakage                  |

| Column Dropped        | Reason                                                                   |
|-----------------------|--------------------------------------------------------------------------|
| paid_error            | Audit flag derived after submission — future leakage                     |
| date_error            | Audit flag derived after submission — future leakage                     |
| is_pending            | Perfectly maps to claim_status=Pending — pure leakage (100% correlation) |
| admission_date        | Raw date replaced by engineered submission_month and is_peak_month       |
| discharge_date        | Raw date — no direct signal; length_of_stay_days captures the duration   |
| claim_submission_date | Raw date replaced by submission_month (integer) and is_peak_month flag   |

## Engineered Features (New Columns Created)

| Engineered Column    | Description                                                                   |
|----------------------|-------------------------------------------------------------------------------|
| is_denied            | Target variable — 1 if claim_status=Denied, else 0                            |
| submission_month     | Integer month extracted from claim_submission_date (1=Jan, 12=Dec)            |
| is_peak_month        | 1 if month is March or August — highest-volume denial months                  |
| is_high_value        | 1 if billed_amount exceeds the dataset median (\$2,924.48)                    |
| provider_denial_rate | Each provider's historical denial rate — captures the Dr. Okafor signal       |
| denial_probability   | Model output — predicted probability of denial (0.0 to 1.0)                   |
| high_denial_risk     | 1 if denial_probability >= 0.5 — the actionable pre-submission flag           |
| confusion_quadrant   | TP / TN / FP / FN label per claim — used for Power BI matrix colouring        |
| data_split           | Train or Test tag — used to filter Power BI confusion matrix to test set only |

## 4. Why ICD Description Was Added

The icd10\_description column was added to the dataset before modelling as a deliberate feature enrichment decision. The original dataset contained icd10\_primary — a standardised alphanumeric diagnosis code (e.g. A09, E11, G43). While these codes are valid predictors, they are opaque: the model encodes them as integers, and the resulting feature importance chart shows an integer rather than a meaningful label.

Adding icd10\_description (the plain-English diagnosis name, e.g. 'Gastroenteritis', 'Type 2 Diabetes Mellitus', 'Viral Infection') provides two key benefits:

- **Richer signal for the model:** The description encodes semantic groupings that the raw code does not — for example, all viral infection codes share vocabulary in their description that the model can leverage through label encoding.
- **Interpretable feature importance:** When the feature importance chart is generated, 'icd10\_description' appears as a recognisable label rather than an encoded integer. This is critical for presenting findings to clinical and finance stakeholders who need to understand WHICH diagnoses are driving denial risk — not

just that 'feature\_7' is important.

The final model was visualised and submitted with icd10\_description included. Comparative testing (see Analysis Report) confirmed that including icd10\_description marginally improved precision (59.57% vs 57.14%) while the AUC remained stable at 0.5760, validating the inclusion decision.

## 5. Model Architecture

| Parameter          | Value                                                                 |
|--------------------|-----------------------------------------------------------------------|
| Algorithm          | Logistic Regression                                                   |
| Solver             | lbfgs (Limited-memory Broyden–Fletcher–Goldfarb–Shanno)               |
| Max Iterations     | 1,000                                                                 |
| Class Weight       | balanced — compensates for the 40/60 class imbalance                  |
| Random State       | 42 (fixed for reproducibility)                                        |
| Scaling            | StandardScaler applied to 6 continuous columns                        |
| Encoding           | One-hot (department, payer_type) + Label (provider, ICD, CPT columns) |
| Train/Test Split   | 80% / 20% — stratified on is_denied to preserve class ratio           |
| Decision Threshold | 0.5 — claims above this score are flagged as high-risk                |

Logistic Regression was selected as the modelling algorithm because it is inherently interpretable — each feature receives a coefficient that directly indicates its direction and magnitude of influence on denial probability. This transparency is essential for a healthcare billing context where stakeholders must understand and trust the model's reasoning. It also produces native probability scores (not just binary predictions), enabling the tiered risk flagging system used in the Power BI dashboard.

## 6. Final Model Performance (With ICD Description)

|                                   |                                  |                                 |                                    |                                   |
|-----------------------------------|----------------------------------|---------------------------------|------------------------------------|-----------------------------------|
| 63.98%<br><small>Accuracy</small> | 0.5760<br><small>ROC-AUC</small> | 28.13%<br><small>Recall</small> | 59.57%<br><small>Precision</small> | 0.3821<br><small>F1 Score</small> |
|-----------------------------------|----------------------------------|---------------------------------|------------------------------------|-----------------------------------|

### Confusion Matrix — Test Set (1,230 Claims)

|                    | Predicted: Not Denied                              | Predicted: Denied                   |
|--------------------|----------------------------------------------------|-------------------------------------|
| Actual: Not Denied | True Negative 650 Correctly cleared                | False Positive 93 Extra review      |
| Actual: Denied     | False Negative 350 Missed denial<br>FINANCIAL RISK | True Positive 137 Correctly flagged |

TN: 650 correctly cleared | FP: 93 false alarms | FN: 350 missed denials | TP: 137 correctly flagged

| Financial Metric                     | Value          |
|--------------------------------------|----------------|
| Average Claim Value                  | \$2,927.44     |
| High-Risk Claims Flagged (all 6,150) | 1,237          |
| Total Value of High-Risk Claims      | \$3,683,942.08 |
| Estimated Leakage — FN (350 claims)  | \$1,040,539.16 |
| Value of Caught Denials — TP (137)   | \$401,059.28   |

## 7. Model Outputs Delivered

| File                         | Contents                                                                                   |
|------------------------------|--------------------------------------------------------------------------------------------|
| novacare_predictions.csv     | All 6,150 claims with denial_probability, high_denial_risk, confusion_quadrant, data_split |
| feature_importance.csv       | All model features ranked by absolute coefficient value                                    |
| model_kpis.csv               | Key performance metrics for Power BI KPI cards                                             |
| confusion_matrix_final.png   | Colour-coded 2x2 confusion matrix chart (test set)                                         |
| roc_curve.png                | ROC curve showing model discrimination ability                                             |
| feature_importance.png       | Top 10 feature importance horizontal bar chart                                             |
| probability_distribution.png | Denial probability score histogram by actual outcome                                       |
| dept_high_risk.png           | % high-risk claims by department bar chart                                                 |
| novacare_lr_model.pkl        | Saved trained model for future claim scoring                                               |
| novacare_scaler.pkl          | Saved StandardScaler for consistent future preprocessing                                   |